

Math 113 – Abstract Algebra

Homework 5 — my solutions

Fall 2026 ■ Keshav Ramamurthy

All assignments must be **turned in on Gradescope** by the deadline. Late assignments are not accepted.

Have questions? Work with your peers, attend office hours, check out the [Student Learning Center](#), and ask in discussion section!

Problems for credit. Turn in these problems; some subset of them will be graded.

1. Fraleigh Exercise 10.6
2. Fraleigh Exercise 10.26
3. Fraleigh Exercise 10.34
4. Fraleigh Exercise 10.37
5. Fraleigh Exercise 11.6
6. Fraleigh Exercise 11.16
7. Fraleigh Exercise 11.50

Practice problems. The following problems will not be graded, and should not be turned in, but are recommended as good practice problems. Solutions to most odd-numbered exercises are available in the back of the textbook.

- Fraleigh Exercises 10.1, 10.3, 10.7, 10.19, 10.27, 10.29, 10.43, 10.44
- Fraleigh Exercises 11.10, 11.26, 11.35, 11.46, 11.51

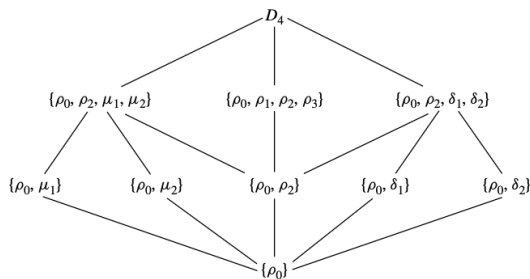
Solutions

Problem 1 (Fraleigh Exercise 10.6)

Solution.

8.12 Table

	ρ_0	ρ_1	ρ_2	ρ_3	μ_1	μ_2	δ_1	δ_2
ρ_0	ρ_0	ρ_1	ρ_2	ρ_3	μ_1	μ_2	δ_1	δ_2
ρ_1	ρ_1	ρ_2	ρ_3	ρ_0	δ_1	δ_2	μ_2	μ_1
ρ_2	ρ_2	ρ_3	ρ_0	ρ_1	μ_2	μ_1	δ_2	δ_1
ρ_3	ρ_3	ρ_0	ρ_1	ρ_2	δ_2	δ_1	μ_1	μ_2
μ_1	μ_1	δ_2	μ_2	δ_1	ρ_0	ρ_2	ρ_3	ρ_1
μ_2	μ_2	δ_1	μ_1	δ_2	ρ_2	ρ_0	ρ_1	ρ_3
δ_1	δ_1	μ_1	δ_2	μ_2	ρ_1	ρ_3	ρ_0	ρ_2
δ_2	δ_2	μ_2	δ_1	μ_1	ρ_3	ρ_1	ρ_2	ρ_0



8.13 Figure Subgroup diagram for D_4 .

ρ_0 is the identity and we are looking at the subgroup $\{\rho_0, \mu_2\}$

The left cosets are of the form $\{g\rho_0, g\mu_2\}$ So, going down the list we have

$$\rho_0(H) = \{\rho_0, \mu_2\}$$

$$\rho_1(H) = \{\rho_1\rho_0, \rho_1\mu_2\} = \{\rho_1, \delta_2\}$$

going off the table. Then

$$\rho_2(H) = \{\rho_2, \mu_1\}$$

and

$$\rho_3(H) = \{\rho_3, \delta_1\}$$

which is a partition of G .

□

Problem 2 (Fraleigh Exercise 10.26)

Solution. The equivalent relation is $a \sim_R b \iff ab^{-1} \in H$.

We have to show reflexive, transitive, and symmetry.

Reflexive: For $a \in G$, $aa^{-1} = e \in H \rightarrow aa^{-1} \in H$. So we have that $a \sim_R a$.

To show Symmetric, let's start from $a \sim_R b$

Then, there exists $ab^{-1} \in H$ and from closure under inverses, we know that $(ab^{-1})^{-1} \in H$.

Simplifying, this is

$$(ab^{-1})^{-1} = (b^{-1})^{-1}a^{-1} = ba^{-1} \iff b \sim_R a$$

To show the transitive property, Let's start with $a \sim_R b$ and $b \sim_R c$.

We have that $ab^{-1} \in H$ and $bc^{-1} \in H$ and since multiplication is closed, we have

$$(ab^{-1})(bc^{-1}) = a(b^{-1}b)c^{-1} = ac^{-1} \iff a \sim_R c$$

□

Problem 3 (Fraleigh Exercise 10.34)

Solution. Given a group of order pq for prime p and q , we want to show that every proper subgroup of cyclic. $|G| = pq$ so by lagrange $|H| \mid |G| \rightarrow |H| \in \{1, p, q\}$

Claim. Every group with prime order is cyclic.

Proof. Consider an element $h \in H, h \neq e$.

By lagrange, the order of the subgroup $\langle h \rangle$ must divide $|H| = p$ so it is either 1 or $|H| = p$.

The $\langle h \rangle = 1$ case would be $\langle e \rangle$, but since $h \neq e$, we have that $|H| = p$ and $\langle h \rangle = H$ and H is cyclic. □

Also, if the subgroup of the group with order pq had size 1, it would be generated again by $\langle e \rangle$ and thus be cyclic.

Thus every proper subgroup must be cyclic. □

Problem 4 (Fraleigh Exercise 10.37)

Solution. Consider an arbitrary group G with no proper non trivial subgroups.

Take $g \in G, g \neq e$ and let's examine $\langle g \rangle$.

Since there are no proper non trivial subgroups, we know that $\langle g \rangle = G$ and G is cyclic.

If G was an infinite cyclic group, notice that $\langle g^2 \rangle = \{\dots, g^{-4}, g^{-2}, e, g^2, g^4, \dots\}$ is a proper non trivial subgroup, and elements like g^{2k+1} are not in this subgroup.

Thus this is a contradiction and G must be finite.

If we suppose for sake of contradiction that $|G|$ is composite, and is equal to $|G| = pq, p \neq 1, q \neq 1$ for some values p and q , we see for example that $\langle g^p \rangle$ is a proper non trivial subgroup (and its order is $\frac{n}{p} = q < |G|$)

Thus we have a contradiction and G must be a cyclic group of prime order. $\square$

Problem 5 (Fraleigh Exercise 11.6)

Solution. Doing them separately, for the order of 3 in $\mathbb{Z}_4$, it would be $\frac{4}{\gcd(3,4)} = 4$

For 10 in $\mathbb{Z}_{12}$, it would be $\frac{12}{\gcd(10,12)} = 6$

Lastly, for 9 in $\mathbb{Z}_{15}$ it would be $\frac{15}{\gcd(9,15)} = 5$. Its a direct product, Taking the least common multiple of these we get

$$\text{lcm}(4, 6, 5) = 60$$

$\square$

Problem 6 (Fraleigh Exercise 11.16)

Solution. We claim that $\mathbb{Z}_2 \times \mathbb{Z}_{12}$ and $\mathbb{Z}_4 \times \mathbb{Z}_6$ are isomorphic. Since $12 = 4 \cdot 3$ and 3 and 4 are relatively prime, $\mathbb{Z}_{12} \cong \mathbb{Z}_3 \times \mathbb{Z}_4$ Thus,

$$\mathbb{Z}_2 \times \mathbb{Z}_{12} \cong \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_4$$

Similarly using the fact that $6 = 2 \cdot 3$ and 2 and 3 relatively prime, we have that:

$$\mathbb{Z}_4 \times \mathbb{Z}_6 \cong \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_2$$

Since changing the orders preserve the isomorphism, we have that

$$\mathbb{Z}_4 \times \mathbb{Z}_6 \cong \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_2 \cong \mathbb{Z}_2 \mathbb{Z}_3 \mathbb{Z}_4$$

$$\mathbb{Z}_2 \times \mathbb{Z}_{12} \cong \mathbb{Z}_4 \times \mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_4$$

as desired. $\square$

Problem 7 (Fraleigh Exercise 11.50)

Solution. We have that $G = H \times K$ and notice an element $(h, k) \in G$ can be written as

$$(h, k) = (h, e_K)(e_H, k) \text{ and } (h, e_K) \in H \times \{e\}, (e_H, k) \in \{e\} \times K$$

So every element in G can be written as hk for $h \in H, k \in K$ and we've shown (a).

For (b), we wish to show $hk = kh$ for $h \in H, k \in K$, so consider arbitrary $(h, e_K) \in H$ and $(e_H, k) \in K$, we have that

$$hk = (h, e_K)(e_H, k) = (he_H, e_Kk) = (h, k)$$

and

$$kh = (e_H, k)(h, e_K) = (e_Hh, ke_K) = (h, k)$$

and thus $hk = kh$ so we've shown part (b).

To show part c which is $H \cap K = \{e\}$.

To show this, consider an element $a \in H \cap K$. Then

$$a = (h, e_K) = (e_H, k) \rightarrow h = e_H, e_K = k \text{ so } a = (e_H, e_K) = e_G$$

and thus $H \cap K = \{e\}$

□